Slope and Rate of Change: Find the Error
Answer Key
Grade 8 / Pre-Algebra

Quick Answers

1. m = 2
2. (a) correct slope = −2, —
3. (a) correct slope = −2, —
4. (a) change in gallons = −18, (b) rate of change = −3, —
5. (a) correct slope = 25, —
6. (a) correct slope = −1, —
7. (a) extra cost in dollars = 36, (b) slope = 6, —
8. (a) slope = −2, —

What is verified

1 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.B.5 – *problems 1, 2, 3, 4, 5, 6, 8*
8.EE.B.6 – *problem 7*
Difficulty 1–4 of 5 across 17 tagged checks.

Common wrong answers

1. *If they got 0.5: inverted the ratio: put the change in x on top*
2. *If they got* −0.5*: inverted the ratio: change in x over change in y*
3. *If they got 2: subtracted the x-values in the opposite order from the y-values, reversing the sign*
4. *If they got* −0.3333*: inverted the ratio: divided the change in hours by the change in gallons*
5. *If they got 0.04: inverted the rate: minutes per page instead of pages per minute*
6. *If they got 1: took the y-difference in one order and the x-difference in the other, losing the negative*
7. *If they got 0.1667: inverted the ratio: divided the change in tickets by the change in cost*
8. *If they got 2: reversed the order of subtraction on the top only, flipping the sign*

1. Find the slope through $A(2, 1)$ and $B(6, 9)$.

Solution: Take both differences from A to B: the change in y is $9 - 1 = 8$ and the change in x is $6 - 2 = 4$. Slope is the y change over the x change, so $m = 8/4$. $\boxed{m = 2}$

2. Jonah: slope through $P(-3,4)$ and $Q(1,-4)$; he wrote $m = \dfrac{1-(-3)}{-4-4} = -\dfrac{1}{2}$. (a) Correct slope. (b) Name his error.

Solution (a): Change in y from P to Q is $-4-4=-8$; change in x is $1-(-3)=4$. So $m=-8/4$. $\boxed{m=-2}$

Solution (b) — instructor-judged. Jonah's two differences are the right two numbers in the wrong places: he wrote the x change on top and the y change underneath, so his fraction is the reciprocal of the slope. A correct response names the inversion ("run over rise instead of rise over run"). "He subtracted wrong" is not enough — nothing was subtracted wrongly.

3. Mara: slope through $R(5,2)$ and $S(1,10)$; she wrote $m = \dfrac{10-2}{5-1} = 2$. (a) Correct slope. (b) Name her error.

Solution (a): Going from R to S, the change in y is $10-2=8$ and the change in x is $1-5=-4$. So $m=8/(-4)$. $\boxed{m=-2}$

Solution (b) — instructor-judged. Mara subtracted in opposite orders: S minus R on top, R minus S on the bottom. Reversing one difference and not the other flips the sign. A correct response says the subtraction must start from the same point in both differences.

4. Draining tank; hours 0, 2, 4, 6 against gallons 40, 34, 28, 22. Ravi says the rate is $-\dfrac{6}{18}$ gallons per hour. (a) Gallons lost from hour 0 to hour 6. (b) Rate of change. (c) Name the error.

Solution (a): Read the two ends of the table: at hour 6 there are 22 gallons and at hour 0 there were 40, so the change is $22-40$. $\boxed{-18}$

Solution (b): The rate of change is gallons per hour, so the gallon change goes on top: $m = \dfrac{-18}{6-0}$. $\boxed{m=-3}$

Solution (c) — instructor-judged. Ravi divided the hour change by the gallon change, so his ratio is upside down; his units come out as hours per gallon, not gallons per hour. A correct response names the inversion, and a units check is full credit on its own.

5. A printer prints 300 pages in 12 minutes. Sam wrote $m = \dfrac{12}{300} = 0.04$. (a) Correct slope in pages per minute. (b) Name his error.

Solution (a): Pages against minutes means pages on top: $m = \dfrac{300-0}{12-0}$. $\boxed{m=25}$

Solution (b) — instructor-judged. Sam divided minutes by pages, which answers a different question — minutes per page. A correct response says pages belong on top because the slope asked for is pages per minute; a units check earns full credit.

6. Ramp through $(1,9)$ and $(7,3)$. Aisha wrote $m = \dfrac{9-3}{7-1} = 1$. (a) Correct slope. (b) Name her error.

Solution (a): From the first point to the second, y goes $3-9=-6$ while x goes $7-1=6$, so $m=-6/6$. $\boxed{m=-1}$

Solution (b) — instructor-judged. Aisha took the y difference starting at the second point but the x difference starting at the first, so the sign flipped. A correct response names the mismatched order; noting that the ramp falls from left to right, so the slope must be negative, is a valid supporting argument.

7. Ticket table: 2, 4, 6, 8 tickets against 15, 27, 39, 51 dollars. Leo says the slope is $\frac{6}{36}$ dollars per ticket. (a) Extra cost of 8 tickets over 2. (b) Slope. (c) Name the error.

Solution (a): From the table, 51 − 15 dollars. $\boxed{36}$

Solution (b): Dollars per ticket puts the dollar change on top: $m = \dfrac{36}{8-2}$. $\boxed{m = 6}$

Solution (c) — instructor-judged. Leo divided the ticket change by the cost change, inverting the ratio; his answer is tickets per dollar. A correct response names the inversion and says the dollar change belongs on top.

8. Dana and Kofi computed the slope through $(-2, 5)$ and $(4, -7)$. Dana got -2; Kofi got 2. (a) Compute it. (b) Explain who is right and what the other did.

Solution (a): y changes by $-7 - 5 = -12$ while x changes by $4 - (-2) = 6$, so $m = -12/6$. $\boxed{m = -2}$

Solution (b) — instructor-judged. Dana is right. Kofi reversed the order on the top only, computing $5 - (-7) = 12$ over the same positive x change, which flips the sign. A correct response identifies Dana and names Kofi's reversed subtraction; the observation that y falls while x rises, so the slope must be negative, is valid supporting evidence.